## CLAIMS

What is claimed is:

**1.** A central processing unit system, comprising:

- a massively parallel optical input/output (I/O) unit integrated on a semiconductor die; and
- an execution core integrated on the same semiconductor die and directly coupled to the optical I/O unit;
- wherein the execution core comprises at least one register and at least one arithmetic logic unit (ALU) configured to operate on data words with a bit-width greater than 128 bits.

**2.** The system of claim 1, wherein the bit-width is 256 bits or greater.

**3.** The system of claim 1, wherein the bit-width is 1024 bits or greater.

**4.** The system of claim 1, wherein the system is configured to receive a data word from the optical I/O unit and load it into the register in a single clock cycle, without serial-to-parallel conversion.

**5.** The system of claim 1, wherein the optical I/O unit is an implementation of the interface described in U.S. Patent Application No. 19/559,705.